

SJSU M179 Midterm

Problem: Prove that any subgraph of $K_{n,n}$ with at least

$$n^2 - n + 1$$

edges has a perfect matching.

Solution:

Proof. Let the two parts of the bipartite graph be L and R , where

$$|L| = |R| = n.$$

Since $K_{n,n}$ has n^2 edges, the graph is missing at most

$$n^2 - (n^2 - n + 1) = n - 1$$

edges.

We prove Hall's condition on L . Suppose, for contradiction, that Hall's condition fails. Then there exists some $S \subseteq L$ such that

$$|N(S)| < |S|.$$

Let $|S| = s$. Then

$$|N(S)| \leq s - 1.$$

So there are at least

$$n - (s - 1) = n - s + 1$$

vertices in R not adjacent to any vertex of S .

Thus every edge from S to these $n - s + 1$ vertices is missing. Hence the graph is missing at least

$$s(n - s + 1)$$

edges. But

$$s(n - s + 1) = n + (s - 1)(n - s) \geq n,$$

since $1 \leq s \leq n$.

Therefore, if Hall's condition failed, the graph would be missing at least n edges. This contradicts that the graph is missing at most $n - 1$ edges.

Hence Hall's condition holds, so there is a matching covering all vertices of L . Since

$$|L| = |R| = n,$$

this matching is a perfect matching. □